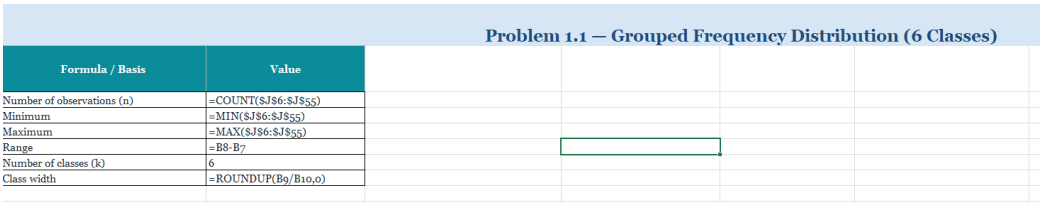
ITC C503 DATA ANALYTICS MIDTERM PROJECT
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MIDTERM PROJECT

Gattoc, Mika Mclaren O.

PROBLEM 1 — GROUPED FREQUENCY DISTRIBUTION

The screenshot below shows the grouped frequency table with lower limits, upper limits, class boundaries, class midpoint/median, frequency, cumulative frequency, and frequency percentage.



Problem 1.1 — Grouped Frequency Distribution (6 Classes)

Formula / Basis	Value
Number of observations (n)	=COUNT($J$6:$J$55)
Minimum	=MIN($J$6:$J$55)
Maximum	=MAX($J$6:$J$55)
Range	=B8-B7
Number of classes (k)	6
Class width	=ROUNDUP(B9/B10,0)

Formula / Computation	**Purpose / Result**
n = COUNT(data)	50 observations
Minimum = MIN(data)	3
Maximum = MAX(data)	254
Range = Maximum − Minimum Number of classes Class width = ROUNDUP(Range / 6, 0)	251 6 42
Lower limit → next class	Previous upper limit + 1
Upper limit	Lower limit + class width − 1
Lower boundary	Lower limit − 0.5
Upper boundary	Upper limit + 0.5
Class midpoint / median	(Lower Limit + Upper Limit) / 2

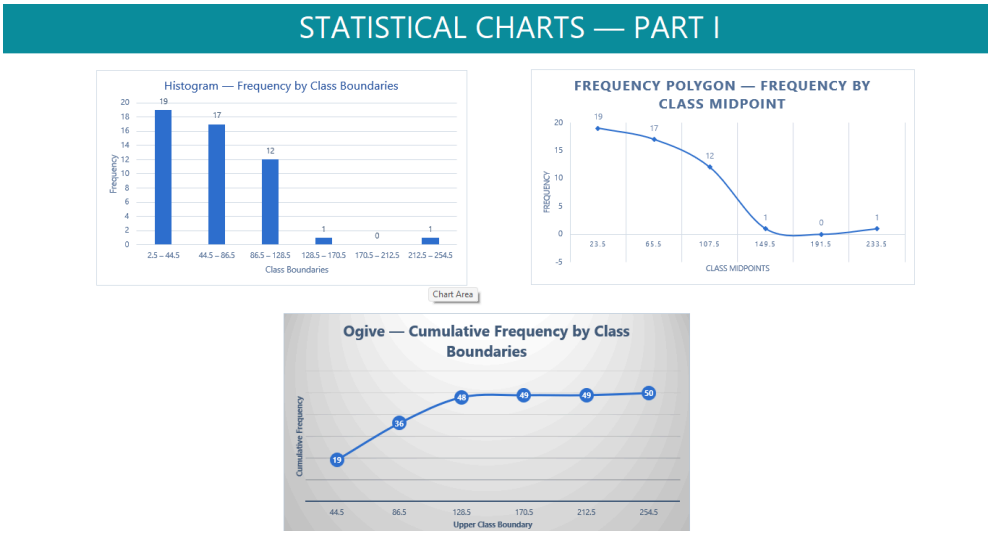
Frequency	COUNTIFS(data, ">="&Lower Limit, data,

	"<="&Upper Limit)
Cumulative frequency	Running total of frequency
Frequency %	Frequency / 50

Final checks: Total Frequency = 50; Total Frequency Percentage = 100%.

PROBLEM 2 — STATISTICAL CHARTS

The three charts below are based on the completed grouped frequency distribution, as required: Histogram, Frequency Polygon, and Ogive.



Formula / Computation	**Purpose / Result**
Histogram	Y-axis = Frequency; X-axis = Class Boundaries
Frequency Polygon	Y-axis = Frequency; X-axis = Class Midpoints

PROBLEM 3 — TOTAL REVENUE BY PRODUCT

Part II spreadsheet summary and corresponding chart.

1. Total Revenue by Product		
Product	Total Revenue	Formula Used
24-inch Monitor	59,288.64	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
27-inch Monitor	85,338.73	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
All-in-One PC	90,256.50	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Desk Organizer	3,417.56	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Desktop Mini	59,915.34	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Ethernet Cable	1,910.55	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
External HDD 2TB	39,044.03	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
External SSD 1TB	62,938.22	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Gaming Laptop	256,377.66	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Keyboard	7,494.75	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Laptop Air 13	157,004.00	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Laptop Pro 14	135,852.48	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Laptop Stand	5,483.65	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Memory Card 256GB	11,692.48	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Mesh Wi-Fi Set	100,250.03	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Mouse Pad	1,857.70	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Network Switch	27,246.76	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Notebook Set	1,496.70	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Pen Pack	638.31	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Portable Monitor	34,395.55	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Printer Paper	1,212.38	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Sticky Notes	875.49	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
USB Drive 128GB	7,633.10	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
USB-C Hub	6,083.40	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Ultrawide Monitor	99,475.05	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Webcam	10,082.26	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Wi-Fi Router	28,431.73	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)
Wireless Mouse	5,624.51	'=SUMIF(Sales_Data!$G$2:$G$1050,A{i},Sales_Data!$L$2:$L$1050)

PROBLEM 4 — AVERAGE ORDERS BY PRODUCT CATEGORY

Part II spreadsheet summary and corresponding chart.

2. Average Orders by Product Category		
Product Category	Average Orders	Formula Used
Accessories	3.17	'=AVERAGEIF(Sales_Data!$F$2:$F$1050,A{i},Sales_Data!$I$2:$I$1050)
Computers	1.42	'=AVERAGEIF(Sales_Data!$F$2:$F$1050,A{i},Sales_Data!$I$2:$I$1050)
Monitors	1.78	'=AVERAGEIF(Sales_Data!$F$2:$F$1050,A{i},Sales_Data!$I$2:$I$1050)
Networking	3.95	'=AVERAGEIF(Sales_Data!$F$2:$F$1050,A{i},Sales_Data!$I$2:$I$1050)
Office Supplies	3.42	'=AVERAGEIF(Sales_Data!$F$2:$F$1050,A{i},Sales_Data!$I$2:$I$1050)
Storage	3.19	'=AVERAGEIF(Sales_Data!$F$2:$F$1050,A{i},Sales_Data!$I$2:$I$1050)

PROBLEM 5 — TOTAL QUANTITY SOLD BY CUSTOMER TYPE

Part II spreadsheet summary and corresponding chart.

3. Total Quantity Sold by Customer Type			
Customer Type	**Total Quantity**	**Formula Used**	
New	929	'=SUMIF(Sales_Data!$E$2:$E$1050,A{i},Sales_Data! Chart Area $1050)	
Returning	1,961	'=SUMIF(Sales_Data!$E$2:$E$1050,A{i},Sales_Data!$I$2:$I$1050)	

PROBLEM 6 — TOTAL REVENUE BY MONTH

Part II spreadsheet summary and corresponding chart.

Formula / Computation	**Purpose / Result**
Formula	SUMIFS(Revenue, Date, first day of month, Date, nex month)
Final result	Total annual revenue/check = 1,301,317.5586.

PROBLEM 7 — LOWEST REVENUE BY REGION

5. Lowest Revenue by Region		
Region	**Lowest Revenue**	**Formula Used**
Central	12.75	'=MIN(IF(Sales_Data!$C$2:$C$1050=A72,Sales_Data!$L$2:$L$1050))
East	6.21	'=MIN(IF(Sales_Data!$C$2:$C$1050=A73,Sales_Data!$L$2:$L$1050))
North	6.46	'=MIN(IF(Sales_Data!$C$2:$C$1050=A74,Sales_Data!$L$2:$L$1050))
South	8.07	'=MIN(IF(Sales_Data!$C$2:$C$1050=A75,Sales_Data!$L$2:$L$1050))
West	6.46	'=MIN(IF(Sales_Data!$C$2:$C$1050=A76,Sales_Data!$L$2:$L$1050))

Chart Area

PROBLEM 8 — HIGHEST REVENUE BY SALES CHANNEL

Part II spreadsheet summary and corresponding chart.

6. Highest Revenue by Sales Channel

Sales Channel	Highest Revenue	Formula Used
Corporate Sales	16,074.24	'=MAX(IF(Sales_Data!$D$2:$D$1050=A82,Sales_Data!$L$2:$L$1050))
Marketplace	6,693.00	'=MAX(IF(Sales_Data!$D$2:$D$1050=A83,Sales_Data!$L$2:$L$1050))
Online	14,85	Chart Area X(IF(Sales_Data!$D$2:$D$1050=A84,Sales_Data!$L$2:$L$1050))
Retail Store	15,120.00	'=MAX(IF(Sales_Data!$D$2:$D$1050=A85,Sales_Data!$L$2:$L$1050))